

Calculating the feature effectiveness indicator (FEI)

Feature effectiveness indicator (FEI)

FEI is a pivotal metric in the realm of software feature evaluation, serving as a singular and comprehensive indicator of the effectiveness of deployed user features in a production environment. FEI encapsulates a range of quantitative data to provide a holistic view of how well a software feature is performing within the production environment. It consolidates various performance indicators, such as category weightage, feature effectiveness measure (FEM), as well as absolute and standard deviation that represent user engagement and satisfaction. The FEI is designed to offer a clear and actionable assessment of whether a feature is meeting its intended objectives and aligning with project goals.

Comparison between FSD and FAD

The choice between feature standard deviation (FSD) and feature absolute deviation (FAD) depends on your specific analytical needs and the characteristics of your data. Use FSD when you want to measure the level of variation or fluctuation in a feature's performance. FSD provides insights into the consistency and stability of the feature's effectiveness over time and is particularly useful when you want to assess how stable the feature's performance has been and whether it exhibits less variation, which is often considered desirable. Use FAD when you want to measure how much the feature's effectiveness values deviate from the mean or average value. FAD quantifies the extent to which the feature's effectiveness varies from the average and it's useful for understanding the degree of fluctuation in the feature's performance and how much it deviates from the mean.

In summary, if you're more interested in assessing the variation and stability of a feature's performance, FSD is a suitable choice. On the other hand, if you want to focus on how much the feature's effectiveness deviates from the average and understand the consistency of its performance, FAD is the metric to use. Your choice should align with your specific analysis goals and the type of insights you aim to gain.

Calculation frequency

FEI is a recurring and dynamic assessment that can be calculated at two specific intervals — at the end of every month and invariably at the end of every quarter. This regular evaluation allows for ongoing monitoring of feature performance and helps identify trends and patterns over time. Monthly assessments provide a more granular view, while quarterly evaluations offer a broader

perspective on feature effectiveness.

Levels of FEI

The FEI uses a simple, straightforward and intuitive 'green' and 'red' scale to classify feature effectiveness, providing a quick and user-friendly visual reference for decision-makers.

Green FEI: When a feature attains a green FEI status, it signifies that the feature is performing optimally and effectively. This level indicates that the feature is achieving or surpassing its predefined growth and deviation expectations. In essence, a green FEI reflects a positive trend and successful performance. Features falling under this category are meeting user needs and contributing positively to the project.

$$\text{Green FEI} = (MFEM > GC \wedge QFEM > GC) \wedge ((MAD < DC \wedge QAD < DC) \vee (MSD < DC \wedge QSD < DC))$$

Yellow FEI: A yellow FEI suggests that a feature's performance is acceptable but there is room for improvement. It may indicate that the feature is progressing well, but there are deviations well above the predefined expectations. While not in a crisis, features with a yellow FEI may benefit from further optimisation to ensure they reach their full potential.

$$\text{Yellow FEI} = (MFEM > GC \wedge QFEM > GC) \wedge ((MAD > DC \vee QAD > DC) \vee (MS > DC \vee QSD > DC))$$

Red FEI: A red FEI is a warning sign that a feature is not performing as expected. It indicates that the feature's performance is falling short of predefined growth and deviation standards. Features categorised as 'red' require immediate attention, investigation, and action. This status prompts stakeholders to address issues, gather user feedback, and make necessary improvements to align the feature with its goals.

$$\text{Red FEI} = (MFEM < GC \vee QFEM < GC)$$

Explanation with examples

Here are the assumptions and deviations:

1. CY = Current year | PY = Previous year
2. CY Q1 = January, February, March | CY Q2 = April, May, June
3. For the ease of understanding, across all the examples:
 - a. GC = 15
 - b. DC = 20
 - c. June CLV CMV changes and all other input metrics remain constant.
 - d. Both FSD and FAD are considered as deviations
4. Calculation happens at the end of CY Q2 June.